

**EXPLORING BEHAVIOURAL SYNDROMES AND PERSONALITY
TRAITS IN AMPHIBIAN LARVAE**

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EXPLORING BEHAVIOURAL SYNDROMES AND PERSONALITY TRAITS IN AMPHIBIAN LARVAE

A project report submitted by

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Table of Contents

List of Figures	iv
List of Tables	v
List of Abbreviations	vi
ACKNOWLEDGEMENTS	vii
ABSTRACT	viii
1. INTRODUCTION	1
1.1. Animal Personality and Behavioural Syndromes	1
1.2. Physiological and Genetic Mechanisms Underlying Personality	1
1.3. Amphibians as a Model for Behavioural Research	2
1.4. Knowledge Gaps and the Tropical Context	3
1.5. Study Species and Rationale	4
1.6. Research Objectives	6
1.6.1 Overall Objective:	6
1.6.2 Specific Objectives:	6
1.7 Hypothesis	6
2. MATERIALS AND METHODS	7
2.1 Study Sites	7
Figure 2.1 (a) Eggs of <i>Polypedates cruciger</i> (b) Eggs of <i>Duttaphrynus melanostictus</i>	7
2.2 Maintenance of Tadpole Larvae	7
2.3 Behavioural Experiments	8
Figure 2.4 a) The custom Python script used for the tracking of the tadpoles b) The tadpole displayed with a yellow colour tracking box during the analysis	10
2.3.1 Emergence Test	11

2.3.2 Open Field Test	11
2.3.3 Novel Object Test.....	12
2.3.4 Mirror Test.....	13
2.4 Data Analysis	14
2.5 Ethical Considerations.....	15
3. RESULTS	15
3.1 Repeatability of Behavioural Traits	15
3.2 Correlations of Behavioural Traits.....	18
4. DISCUSSION.....	29
5. REFERENCES	36

List of Figures

Figure 2.1 (a) Eggs of <i>Polypedates cruciger</i> (b) Eggs of <i>Duttaphrynus melanostictus</i>	7
Figure 2.2 Timeline of the short-term experimental procedure emergence test (ET), mirror test (MT), novel object test (NOT), open field test (OFT)	9
Figure 2.3 Timeline of the long-term experimental procedure emergence test (ET), mirror test (MT), novel object test (NOT), open field test (OFT)	9
Figure 2.4 a) The custom Python script used for the tracking of the tadpoles b) The tadpole displayed with a yellow colour tracking box during the analysis.....	10
Figure 2.5 a) Schematic diagram of the arena, showing shelter area with an artificial plant, small stones and a movable door to provide access from the shelter to the rest of the tank. Risky zone was segmented manually b) Photograph during pilot study with <i>Duttaphrynus melanostictus</i> tadpole in the open field area.....	11
Figure 2.6 a) Schematic diagram of the arena used for novel object test. b) and c) Photographs taken from the video recordings of novel object tests with two different objects.....	12
Figure 2.7 a) Schematic diagram of the arena used for mirror test b) Photographs taken from the video recordings of the mirror test.....	13
Figure 3.1 a) Within-individual (above diagonal) and among-individual (below diagonal) correlations between behavioural variables in <i>Polypedates cruciger</i> short-term	23
Figure 3.1 b) Within-individual (above diagonal) and among-individual (below diagonal) correlations between behavioural variables in <i>Polypedates cruciger</i> long- term	23
Figure 3.2 a) Within-individual (above diagonal) and among-individual (below diagonal) correlations between behavioural variables in <i>Duttaphrynus melanostictus</i> short-term	28
Figure 3.2 b) Within-individual (above diagonal) and among-individual (below diagonal) correlations between behavioural variables in <i>Duttaphrynus melanostictus</i> long-term.....	28

List of Tables

Table 2.1 The behavioural traits measured, and the procedures used to measure them in <i>Polypedates cruciger</i> and <i>Duttaphrynus melanostictus</i> larvae	8
Table 3.1 Repeatability estimates (with 95% credible intervals in parentheses) for each personality trait in common hourglass frog (<i>Polypedates cruciger</i>) measured across short-term (n = 50) and long-term time periods (n= 50).....	16
Table 3.2 Repeatability estimates (with 95% credible intervals in parentheses) for each personality trait in common toad (<i>Duttaphrynus melanostictus</i>) measured across short-term (n = 50) and long-term time periods (n = 50).....	17
Table 3.3 Within individual, among individual and overall phenotypic correlations (with 95% credible intervals in parentheses) between behavioral variable in common hourglass frog (<i>Polypedates cruciger</i>) measured across short-term (n = 50) and long-term time periods (n=50)	20
Table 3.4 Within individual, among individual and overall phenotypic correlations (with 95% credible intervals in parentheses) between behavioral variable in common toad (<i>Duttaphrynus melanostictus</i>) measured across short-term (n = 50) and long-term time periods (n=50)	25

List of Abbreviations

ET – emergence test

OFT – open field test

NOT – novel object test

MT – mirror test

TL – track length

RZ – time spent in risky zone

ET – emergence time

LA – latency to approach an object

TS – time spent near object

CV – time spent near mirror

AT – number of attacks towards mirror

CM – latency to approach mirror

Avoid – number of avoidances done away from the mirror

Expo – number of movements done parallel to the mirror.

POLS – Pace of Life Syndrome

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ABSTRACT

Understanding consistent individual differences in behaviour, or animal personality, and their ecological significance is fundamental for interpreting species interactions, competitive dynamics, and developmental trajectories. This study provides the first comprehensive assessment of repeatability (personality) and behavioural syndromes (correlations) in the larval stages of two sympatric Sri Lankan anurans: the endemic *Polypedates cruciger* and the widespread *Duttaphrynus melanostictus*. Using repeated short- and long-term behavioural assays and multivariate mixed-effects models, we quantified repeatability in key behavioural traits including boldness, exploration, sociability, aggression, and activity, and examined inter-trait correlations at phenotypic, within-individual, and among-individual levels to characterize behavioural syndromes. *Polypedates cruciger* exhibited low-to-moderate short-term repeatability but high long-term repeatability across exploration, sociability, and boldness, while activity and risky-zone use remained non-repeatable. Correlation analyses revealed strong positive phenotypic and among-individual associations among exploration and sociability traits, together with within-individual trade-offs between activity and sociability and between boldness and sociability, indicating a structured proactive-like behavioural syndrome. In contrast, *D. melanostictus* displayed exceptionally high short-term repeatability in exploration, sociability, and boldness, which declined over the long term, while aggression increased in repeatability. Correlations were generally weaker and more diffuse but included positive phenotypic, within-individual, and among-individual associations linking exploration, sociability, boldness, and aggression, reflecting flexible yet integrated behavioural syndromes. Activity and risky-zone use were non-repeatable in both species. These species-specific patterns of repeatability (personality) and behavioural syndromes (correlations) correspond with ecological specializations, competitive strategies, and developmental trajectories, providing mechanistic insight into asymmetric competition and establishing essential baseline data for amphibian behavioural ecology, ontogenetic development, and conservation management.

Keywords: Activity, Behaviour, *Duttaphrynus melanostictus*, *Polypedates cruciger*, Repeatability

1. INTRODUCTION

1.1. Animal Personality and Behavioural Syndromes

Over the past several decades, research in animal behaviour has shifted from reliance on population-level averages to recognizing the importance of consistent individual differences. These stable differences in behaviour are now widely conceptualized as animal personality, reflecting repeatable traits that persist over time and across contexts (Wilson et al., 1994; Kelleher et al., 2018). Core personality traits such as boldness, aggression, exploration, activity, and sociability influence how individuals interact with their environment, affect ecological interactions, and contribute to fitness outcomes (Réale et al., 2007; Bell et al., 2009). Unlike transient behavioural responses, these traits represent intrinsic tendencies underpinned by genetic, physiological, and cognitive mechanisms, making them central to behavioural ecology (Wolf and Weissing, 2012; Dingemanse et al., 2010).

A closely related concept, behavioural syndromes, refers to suites of correlated behaviours consistently expressed across multiple contexts (Sih et al., 2004). For example, individuals that are bold in predator encounters may also be aggressive during competition or more exploratory while foraging. These syndromes are maintained through ecological and evolutionary pressures, revealing functional associations among traits and constraining or facilitating behavioural plasticity (Bell et al., 2007; Kelleher et al., 2018). One widely studied example is the proactive-reactive axis, where proactive individuals are typically more active, bold, and exploratory, while reactive individuals are more cautious and flexible (Sih et al., 2004; Koprivnikar et al., 2012; Ferrari et al., 2013). Behavioural syndromes can therefore directly influence survival, growth, reproductive success, and susceptibility to predation or parasitism (Urszán et al., 2015).

1.2. Physiological and Genetic Mechanisms Underlying Personality

Animal personality traits are closely linked to underlying physiological, genetic, and neurobiological mechanisms. Metabolic rate, for instance, has been associated with activity and boldness, while neuroendocrine responses, including glucocorticoid and serotonin pathways, mediate coping strategies and aggression (Biro et al., 2008; Koolhaas et al., 1999). Cognitive traits such as cerebral lateralization and learning ability also contribute to consistent behavioural differences (Reddon et al., 2009). Early developmental experiences, including maternal effects,

resource availability, predator exposure, or parasitic infection, can shape enduring behavioural trajectories, sometimes via epigenetic mechanisms (Crews and McLachlan, 2006; Burlibasa et al., 2011).

Quantifying personality relies on metrics such as repeatability, the proportion of behavioural variance attributable to individual differences (Lessells et al., 1987; Dingemanse et al., 2013). High repeatability supports the presence of distinct personalities, whereas low repeatability indicates plastic or context-dependent behaviour. Across taxa, including mammals, birds, reptiles, fish, and invertebrates, high repeatability of behaviour has been demonstrated, confirming the ecological and evolutionary relevance of personality (Réale et al., 2010; Martins et al., 2012; Pathirana et al., 2025).

1.3. Amphibians as a Model for Behavioural Research

Amphibians are increasingly recognized as valuable models for studying personality and behavioural syndromes due to their unique biphasic life cycles, which include aquatic larval and terrestrial or semi-terrestrial adult stages. This ontogenetic transition provides opportunities to assess behavioural consistency across development and to examine how ecological pressures shape individual differences (Kelleher et al., 2018). Larval amphibians are experimentally tractable: they can be produced in large numbers, are easy to rear under controlled conditions, and respond sensitively to environmental variability (Burlibasa et al., 2005).

Amphibian larvae experience dynamic ecological pressures, including predation, competition, parasitism, and resource fluctuations, making them ideal for studying context-dependent behavioural strategies (Szuroczki and Richardson, 2012; Koprivnikar et al., 2012). Their permeable skin and complex life cycles make them highly sensitive to environmental disturbances, positioning them as effective bio indicators of ecosystem health (Walls and Gabor, 2019). Recent studies have documented repeatable differences in larval boldness, activity, exploration, and social behaviour across species such as *Lithobates sylvaticus*, *Ambystoma maculatum*, *Rana dalmatina*, and bullfrogs (*Lithobates catesbeianus*) (Kelleher et al., 2018; Bégué et al., 2023; Koprivnikar et al., 2012; Adam et al., 2018; Urszán et al., 2015).

1.4. Knowledge Gaps and the Tropical Context

Despite substantial progress in understanding animal personality and behavioural syndromes, important gaps remain in amphibian research, particularly within tropical ecosystems. Most studies on personality traits have been conducted in temperate regions and on a limited set of model species, leaving tropical amphibians often highly diverse and ecologically specialized largely understudied (Kelleher et al., 2018; Koprivnikar et al., 2012). Tropical ecosystems present unique challenges and selective pressures, including high species richness, complex predator-prey networks, intense competition for limited resources, and pronounced temporal environmental variability. These factors may shape behavioural trait expression in ways that differ substantially from temperate systems, highlighting the need for focused research in these regions.

Larval amphibians in tropical environments face a variety of ecological stressors that can influence personality and behavioural syndromes. Predation is a major selective force, with invertebrate predators, fish, and other amphibians exerting pressure on larvae (Relyea, 2001; Szuroczki and Richardson, 2012). The intensity of predation may select for specific behavioural strategies, such as boldness to secure resources versus cautious behaviour to avoid detection. However, how these selective pressures interact with consistent individual differences is not fully understood, particularly in species with contrasting life histories. Similarly, resource distribution and competition can influence activity and exploratory behaviour, with individuals differing in their propensity to exploit novel habitats or food patches. In tropical ponds and streams, resource availability can fluctuate dramatically due to rainfall patterns, seasonal water flow, and nutrient inputs, creating variable ecological contexts in which personality traits are expressed (Koprivnikar et al., 2012; Bégué et al., 2023).

Parasitism and disease add another layer of complexity to larval behavioural ecology. Parasitic infections can modify host behaviour through physiological stress, altered energy allocation, or neurological effects, potentially impacting survival and growth (Pathirana et al., 2025). Despite growing evidence in other vertebrate taxa that parasites can influence personality, few studies have examined these effects in tropical amphibians, leaving a critical gap in understanding the ecological drivers of behavioural variation. Additionally, the interaction between parasitism, predation, and resource competition may generate context-dependent behavioural responses,

emphasizing the need for integrative studies that combine ecological, physiological, and behavioural perspectives.

Climate and environmental variability are particularly salient in tropical regions. Amphibian larvae are highly sensitive to changes in water temperature, pH, and hydrological patterns, with potential consequences for metabolic rate, activity levels, and behavioural syndromes (Walls and Gabor, 2019). Extreme weather events, such as droughts and floods, can alter pond connectivity, larval density, and predation risk, creating shifting selective landscapes for behaviour. Despite these pressures, there is limited empirical research on how consistent individual differences in behaviour are maintained under fluctuating tropical conditions, or how plasticity interacts with intrinsic personality traits to influence fitness outcomes.

Tropical amphibians also display remarkable phenotypic and developmental plasticity, yet the mechanisms linking early-life environmental conditions to adult behavioural traits remain poorly characterized. Epigenetic modifications, maternal effects, and early exposure to stressors can shape the emergence and stability of personality traits, but evidence from tropical systems is scarce (Waddington, 1953; Crews and McLachlan, 2006; Godin et al., 2012). Integrating these developmental perspectives with behavioural assays is essential for understanding the ontogeny of personality and the potential for transgenerational inheritance of behavioural traits.

Sri Lanka, as a tropical biodiversity hotspot with high levels of amphibian endemism, exemplifies both the urgency and opportunity for such research. While taxonomic and conservation assessments are extensive, behavioural ecology remains largely unexplored, limiting our ability to predict species responses to habitat alteration, pollution, invasive species, and climate change (Manamendra-Arachchi and Pethiyagoda, 2006). Investigating personality traits and behavioural syndromes in this context provides critical insights into how individual-level behavioural variation contributes to population resilience, ecological interactions, and long-term persistence in rapidly changing environments.

1.5. Study Species and Rationale

This thesis addresses the identified gaps by systematically examining behavioural syndromes and personality traits in the larvae of two anuran species: common hourglass frog (*Polypedates cruciger*), endemic to Sri Lanka, and Common toad (*Duttaphrynus melanostictus*), and a widespread species across Asia. These species were selected due to their contrasting

reproductive strategies, larval ecologies, and habitat preferences, which provide an ideal framework for comparative behavioural research. *Polypedates cruciger* deposits eggs in foam nests attached to vegetation above water bodies, resulting in larvae that fall into the water upon hatching. This reproductive mode exposes larvae to specific microhabitat conditions, including variation in predation pressure, water chemistry, and resource availability. In contrast, *D. melanostictus* lays eggs as aquatic gelatinous strings directly in the water, producing larvae that are immediately immersed in the aquatic environment, where they experience different spatial distributions, competition levels, and predator interactions (Khan, 1982; Manamendra-Arachchi and Pethiyagoda, 2006).

These differences in reproductive strategy and larval ecology are expected to influence not only the expression of individual behavioural traits but also the structure of behavioural syndromes. For instance, foam-nesting species may favour larvae that exhibit rapid escape responses and high exploration to locate food or avoid predators, whereas larvae from gelatinous strings may display more cautious behaviour due to higher exposure to conspecific competition and predators. By examining these two species in parallel, this study can explore how life-history traits and early ecological experiences shape personality traits and their repeatability over time.

Furthermore, both species occupy overlapping habitats in parts of Sri Lanka, enabling an assessment of intra- and inter-specific variation in response to similar environmental pressures (Ukuwela & Ranawana, 2011). Such sympatric comparisons are valuable for disentangling the relative contributions of species-specific evolutionary strategies and immediate ecological context to behavioural variation. Investigating these differences in larval behaviour is particularly important because larval stages represent a critical period for survival and growth, which ultimately influences metamorphic success, adult fitness, and population dynamics.

From a conservation perspective, *P. cruciger* represents an endemic species highly vulnerable to habitat alteration and climate change, while *D. melanostictus* is widespread and resilient. Understanding how personality traits and behavioural syndromes manifest in these larvae provides insight into potential adaptive strategies that influence survival under environmental stress. This information is critical for conservation management in tropical biodiversity hotspots, where habitat fragmentation, pollution, and invasive species threaten endemic amphibians.

Additionally, the study of larval behaviour in these two species contributes to broader ecological and evolutionary theory. Behavioural syndromes and consistent individual differences have been shown to affect dispersal, resource use, competitive interactions, and susceptibility to predation and parasitism in other taxa. By documenting these patterns in tropical amphibian larvae, this thesis can inform both fundamental behavioural ecology and applied conservation practices, such as habitat restoration, captive breeding programs, and population monitoring.

1.6. Research Objectives

1.6.1 Overall Objective:

To explore behavioural syndromes and personality traits in amphibian larvae (*Polypedates cruciger* and *Duttaphrynus melanostictus*).

1.6.2 Specific Objectives:

1. To assess the repeatability and consistency of behavioural traits activity, boldness, exploration, sociability, and aggression in *P. cruciger* and *D. melanostictus* larvae.
2. To examine intra- and inter-specific correlations between behavioural syndromes and personality traits in *P. cruciger* and *D. melanostictus* larvae.
3. To evaluate short-term and long-term behavioural consistency in *P. cruciger* and *D. melanostictus* larvae through a series of behavioural assays.

1.7 Hypothesis

1. There is a repeatability and consistency of behavioural syndromes and personality traits in selected anuran species.
2. Consistent individual differences in behavioural traits (personality) across species will form behavioural syndromes, and these syndromes will be influenced by both inter- and intra- specific interactions and environmental pressures.
3. Amphibian larvae exhibit consistent individual traits (personality traits) over both short-term (within days) and long-term (across developmental stages) periods.

2. MATERIALS AND METHODS

2.1 Study Sites

Sampling of *Polypedates cruciger* and *Duttaphrynus melanostictus* eggs (Figure 2.1a & b) was conducted at two locations in Kandy, Sri Lanka: the Royal Botanical Garden (Latitude: 7°16' 9.60" N, Longitude: 80°35' 26.39" E) and the University of Peradeniya premises (Latitude: 6.9737° N, Longitude: 79.9153° E). These sites were selected based on the accessibility of breeding populations and the presence of suitable aquatic microhabitats for egg deposition.



(a)



(b)

Figure 2.1 (a) Eggs of *Polypedates cruciger* (b) Eggs of *Duttaphrynus melanostictus*

2.2 Maintenance of Tadpole Larvae

Eggs of both species were collected from their natural habitats and transported to the laboratory in plastic containers filled with water from the collection sites. The eggs were allowed to hatch under laboratory conditions. From each species, fifty tadpoles at ten days post-hatch were selected for experiments. Initially, tadpoles were housed together in a single tank (61 × 31 × 31 cm) for one week. Subsequently, individual tadpoles were transferred to separate isolation tanks (32 × 14 × 14 cm) for three days. Individual housing allowed for identification without the stress of marking and ensured standardized living conditions while preserving visual contact between tadpoles (Frederick, 1997; Ploverino et al., 2016).

Tadpoles were maintained under a 12:12 light-dark photoperiod at 27–29 °C. They were fed 2.0 g of commercial fish food daily, and aged, de-chlorinated tap water was used, replaced weekly. Tadpoles were acclimatized for three days prior to behavioural testing.

2.3 Behavioural Experiments

Five personality traits activity, boldness, exploration, sociability, and aggression were assessed using four behavioural assays: The Open Field Test (OFT), Emergence Test (ET), Novel Object Test (NOT), and Mirror Test (MT) (Table 2.1).

Table 2.1 The behavioural traits measured, and the procedures used to measure them in *Polypedates cruciger* and *Duttaphrynus melanostictus* larvae

Behavioral trait	Putative personality trait	Procedure
Track length (TL)	Activity	Open field test
Time spent in the risky zone (RZ)	Boldness	Open field test
Emergence time (ET)	Boldness	Emergence test
Time spent near the object (TS)	Exploration	Novel object test
Latency to approach the object (LA)	Exploration	Novel object test
Time spent closer to mirror (CV)	Sociability	Mirror test
No of attacks done towards the mirror (AT)	Aggression	Mirror test
Movements done parallel do the mirror (Expo)	Exploration	Mirror test
No of avoidances done away from the mirror (Avoid)	Aggression	Mirror test

Seven behavioural variables corresponding to the personality traits were recorded for both *P. cruciger* and *D. melanostictus* tadpoles. Behavioural assays were conducted in both short-term and long-term experimental designs (Figure 2.2 and Figure 2.3).

Short-term tests were performed across four time points, with a one-day rest between tests and a one-week rest between time points (Figure 2.2). Long-term tests consisted of three or four time points, depending on larval morphology, with one-day rests between tests and a three-week interval between time points (Figure 2.3).

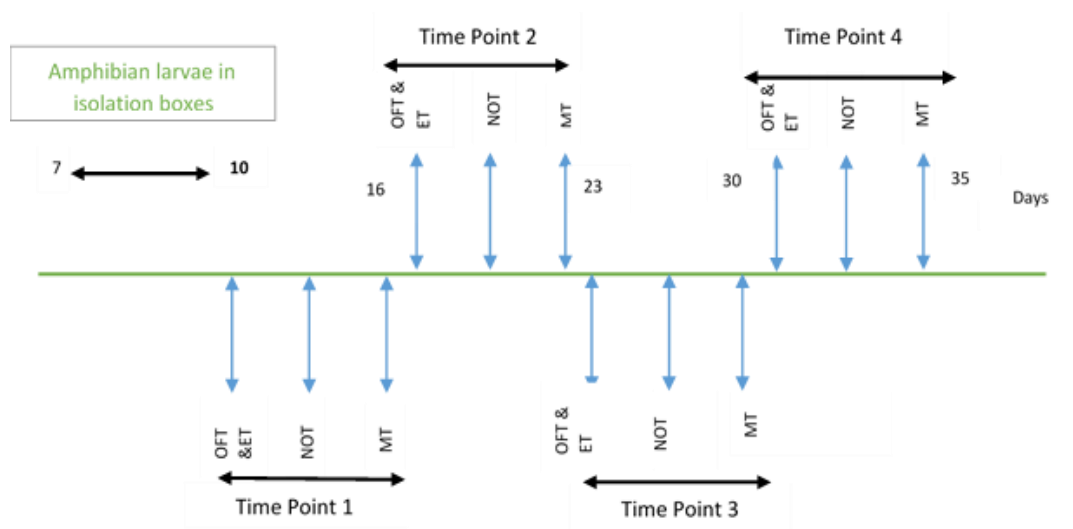


Figure 2.2 Timeline of the short-term experimental procedure emergence test (ET), mirror test (MT), novel object test (NOT), open field test (OFT)

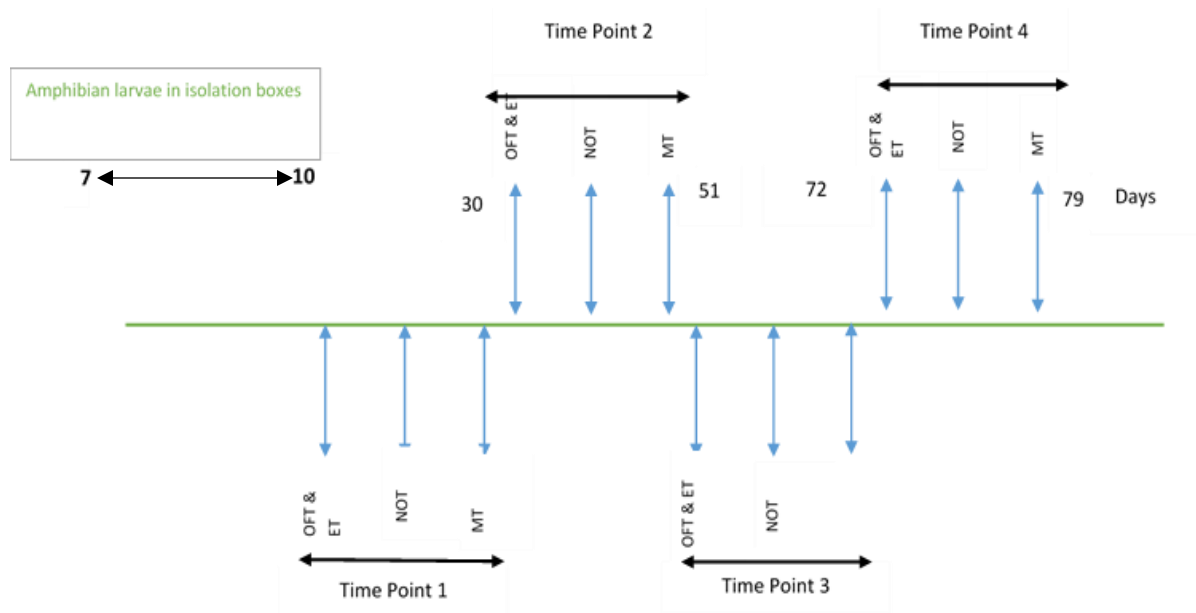


Figure 2.3 Timeline of the long-term experimental procedure emergence test (ET), mirror test (MT), novel object test (NOT), open field test (OFT)

The purpose of the four behavioural tests was to assess multiple personality traits and examine correlations among them. The ET measured latency to emerge from a shelter, reflecting boldness, and allowed assessment of boldness-activity syndromes. The OFT quantified activity

and boldness, as well as activity-exploration and activity-boldness behavioural syndromes. The NOT evaluated exploration and interspecies differences in novelty response. The MT assessed sociability, aggression, exploratory versus avoidance behaviour, and responses to perceived threats, allowing evaluation of intra- and interspecies social responsiveness. All behavioural assays were conducted between 08:00 and 16:00 in tanks ($32 \times 14 \times 14$ cm) filled with 6 cm of water. Tanks were covered with translucent polythene to prevent visual disturbances, and water temperature was maintained at 27–29 °C. Tadpole behaviour was recorded using a high-resolution HD camera (60 fps) mounted 1 m above the tanks. Videos were analysed using a specialized custom Python script developed in collaboration with research partners (Figure 2.4 a & b)

```

1  import os
2  import cv2
3  import numpy as np
4  import cv
5  import time
6
7  # video_filename = 'mirror test perch.avi'
8  # video_filename = 'Tadpoles video_19125.mp4'
9  # video_filename = '14 Tadpole Off.mp4'
10 # video_filename = '15 Tadpole Off.mp4'
11 # video_filename = 'first tadpole Mirror test St .mp4'
12 # video_filename = 'first tadpole Not St .mp4'
13 video_filename = 'first tadpole off .mp4'
14
15 # Get the directory of the current script or executable
16 current_directory = os.path.dirname(os.path.abspath(__file__))
17
18 # Define a relative folder path
19 relative_folder_path = os.path.join(current_directory, 'videos')
20
21 # Define the video location
22 video_location = os.path.join(relative_folder_path, video_filename)
23
24 # Initialize the background subtractor
25 fgbg = cv.createBackgroundSubtractorMOG2(history=500, varThreshold=16, detectShadows=True)
26
27 # Variables to store the last detected position
28 last_x, last_y, last_w, last_h = None, None, None, None
29 no_movement_frames = 10
30 max_no_movement_frames = 10 # Number of frames to keep the bounding box after the fish stops
31
32 # List to store centroid and timestamp data
33 centroid_data = []
34 positions = []
35
36
37 # Function to get the current time in milliseconds
38 def current_time_ms():
39     return int(round(time.time() * 1000))
40
41
42 # Function to format time as hh:mm:ss.ms
43 def format_time(ms):
44     seconds = ms // 1000
45     minutes = seconds // 60
46     hours = minutes // 60
47     milliseconds = ms % 1000
48     return f'{hours:02}:{minutes:02}:{seconds:02}.{milliseconds:03}'
49
50
51 # Record start time
52 start_time = current_time_ms()
53
54

```

(a)



(b)

Figure 2.4 a) The custom Python script used for the tracking of the tadpoles b) The tadpole displayed with a yellow colour tracking box during the analysis.

2.3.1 Emergence Test

Tadpoles were placed in a shelter area containing a small plastic plant and three stones. After a 3-minute acclimatisation, latency to emerge from the shelter (ET) was recorded using a stopwatch, and behaviour was video-recorded for analysis. The emergence of amphibian larvae from the shelter was observed using the camera connected to a computer.

2.3.2 Open Field Test

The shelter door was closed after they came out of the shelter. The behaviour of the amphibian larvae was filmed for 5 minutes using the camera (Figure 2.5 a & b). The track length (TL) and the amount of time spent in the risky zone (RZ) were analysed by using the videos and the software. Risky zone is an area located away from the surfaces, segmented using the customized Python script which was used in the experiment. After this amphibian larvae were captured and returned to their isolation tanks (Figure 2.5 a & b).

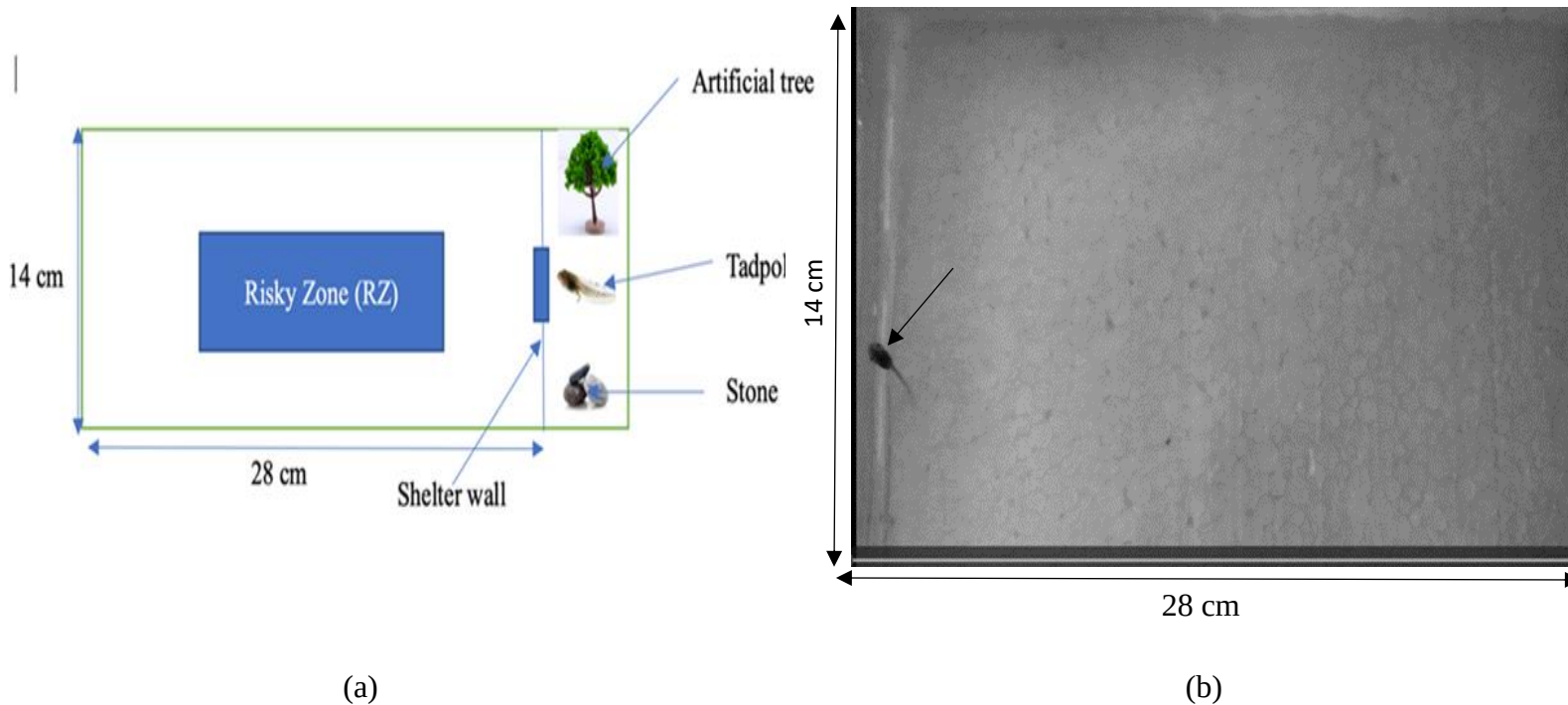


Figure 2.5 a) Schematic diagram of the arena, showing shelter area with an artificial plant, small stones and a movable door to provide access from the shelter to the rest of the tank. Risky zone was segmented manually b) Photograph during pilot study with *Duttaphrynus melanostictus* tadpole in the open field area.

2.3.3 Novel Object Test

Amphibian larvae were taken individually and gently placed in the middle of the experimental tank with dimensions 32 x 14 x 14 cm with water filled up to 6.0 cm (Figure 2.6 a & b and c). The larvae were allowed to acclimatize for 3 minutes. A blue pebble (2.5 cm length x 0.5 cm width) at the time point – 1, a coin (2 cm radius) at the time point – 2, a red piece of Lego (3 cm radius, 5 cm height) at the time point – 3 and a blue, small ball (radius 1.5 cm) at the time point 4 were introduced to the tank and the behaviour of the amphibian larvae were observed for 10 minutes using the camera (Figure 2.6 a & b and c). The analysis of the video was done by using the software to find out the latency to approach the object (LA) and track length (TS). After this amphibian larvae were captured and returned to their isolation tanks.

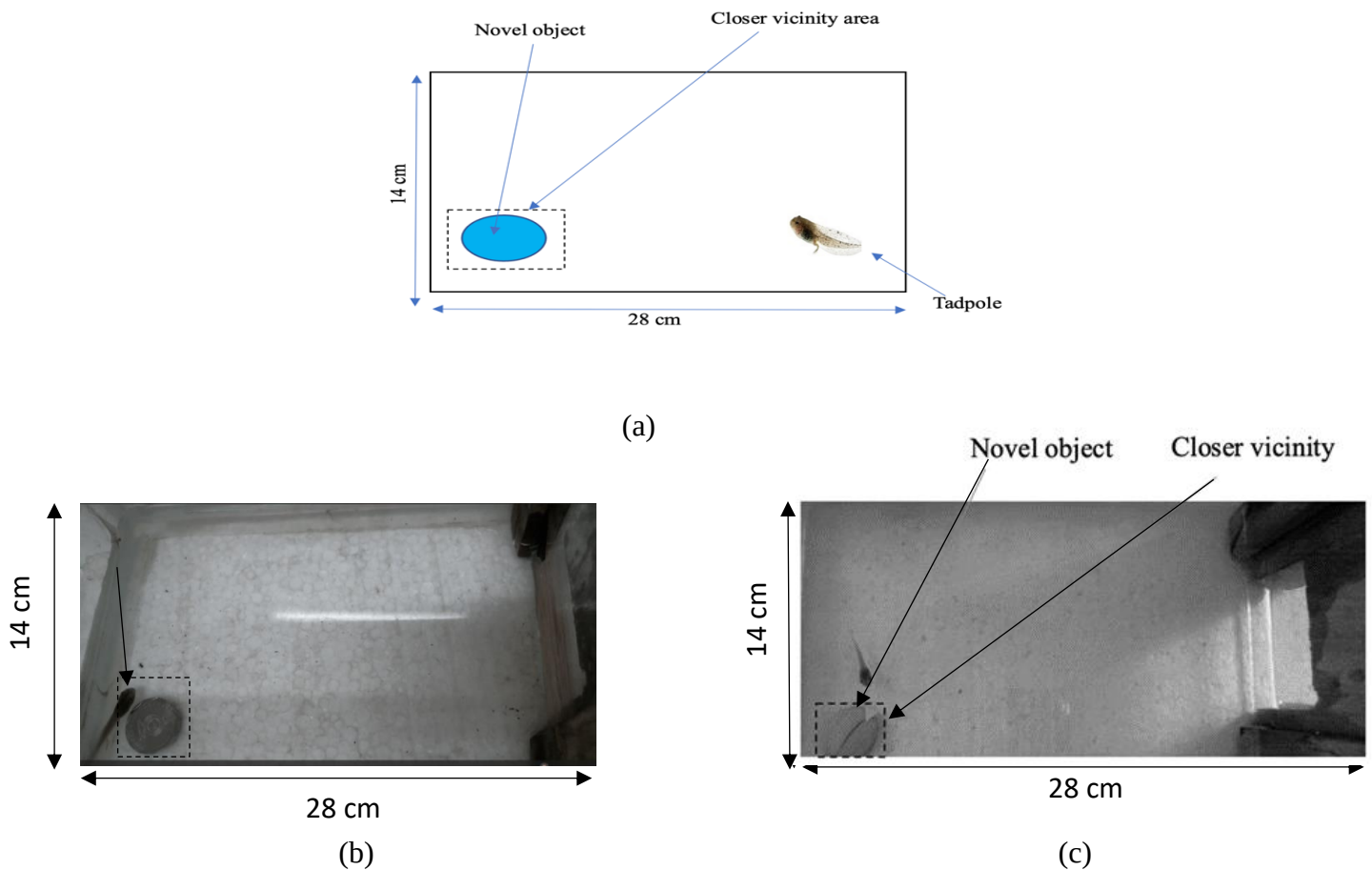
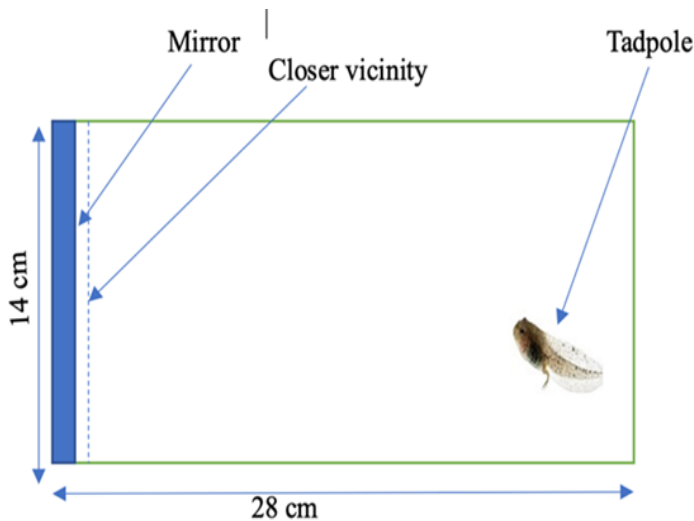


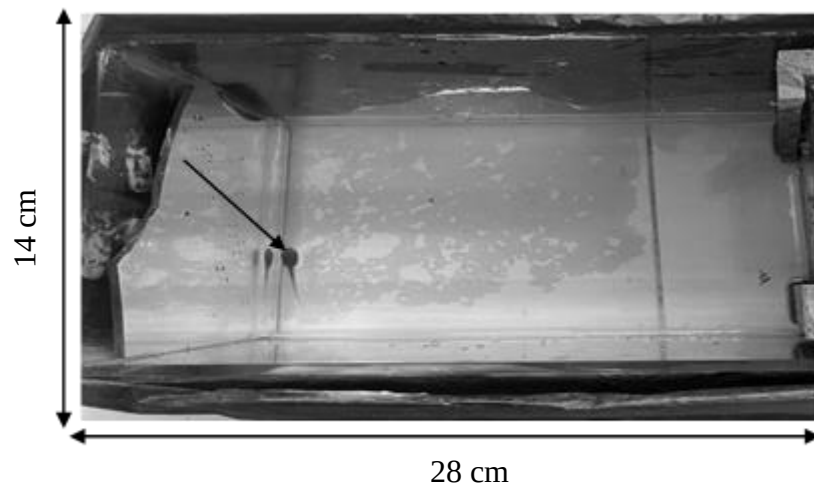
Figure 2.6 a) Schematic diagram of the arena used for novel object test. b) and c) Photographs taken from the video recordings of novel object tests with two different objects.

2.3.4 Mirror Test

Amphibian larvae were taken individually and were placed inside the experimental tank which contains a mirror (12 x 5 x 0.5 cm) fitted to one wall of the tank (Figure 2.7 a & b). The tadpole was placed away from the mirror after letting them to stay 3 min of acclimatization time. The behaviour of the amphibian larva was filmed for 5 minutes using the camera. The videos taken were analysed to measure the time spent closer to mirror (CV), number of attacks towards the mirror (AT), time spent near the mirror zone (CM), movements done parallel do the mirror (Expo) and no of avoidances done away from the mirror (Avoid). Exploration, sociability and aggression were measured. After this amphibian larvae were captured and returned to their isolation tanks. Aggression during the mirror test was quantified by recording the number of aggressive acts directed at the mirror image, including lunges, attacks, or striking movements, along with the latency to the first attack and total duration of aggressive behaviour. Exploration was measured by counting the number of non-aggressive approaches toward the mirror, characterized by slow swimming, close inspection from different angles, gentle touching of the mirror, and swimming parallel to the reflection. Latency to the first exploratory approach and total time spent near the mirror without aggression were also recorded as indicators of information gathering rather than confrontation.



(a)



(b)

Figure 2.7 a) Schematic diagram of the arena used for the mirror test b) Photograph taken from the video recording of mirror test of *Duttaphrynus melanostictus* tadpole

After experiments, tadpoles were euthanized humanely using an immersion dose of MS-222 (tricaine methanesulfonate) at 300 mg/L, buffered to pH 7.0–7.5, following American Veterinary Medical Association (AVMA) and Federation of European Laboratory Animal Science Associations (FELASA) guidelines.

2.4 Data Analysis

All statistical analyses were conducted using R version 3.6.1 (R Core Team, 2021). For each behavioural variable ET, TL, RZ, LA, TS, CV, AT, CM, Avoid and exploration adjusted repeatability (i.e., repeatability estimates after controlling for fixed effects; Nakagawa & Schielzeth, 2010; Pathirana et al., 2025) was calculated by fitting linear mixed-effects models using the R package rptR (Stoffel et al., 2017). Log-transformed ET and TL, as well as RZ, LA, TS, CV, CM, Expo and Avoid were modelled as Gaussian variables, while Attacks was modelled using a Poisson distribution. Individual ID was included as a random effect, with body mass, body length and time point as fixed effects. Repeatability was estimated as the intra-class correlation coefficient (ICC) derived from within-individual and among-individual variance components. Confidence in these estimates was assessed using 1,000 parametric bootstrap replicates to generate 95% confidence intervals (cis). Statistical significance of repeatability was determined by permutation tests of the residuals from the fitted null models, and a trait was considered significantly repeatable only if $p < 0.05$ in the permutation test and the 95% CI did not include zero.

To examine correlations between behavioural traits, within-individual, among-individual, and overall phenotypic correlations were calculated using multivariate mixed-effects models under a Bayesian framework, implemented in the mcmcglmm package (Hadfield, 2010; Dingemanse & Dochtermann, 2013). Response variables and fixed/random effects were modeled as described above. Models were run for 1,500,000 iterations with a burn-in of 500,000 and a thinning interval of 100. Behavioural correlations were considered significant if the 95% credibility interval (CI) did not include zero.

2.5 Ethical Considerations

Ethical approval for all procedures was obtained from the Ethics Committee of the Postgraduate Institute of Science, University of Peradeniya (Permit No: CEC_PGIS_2025_03). All procedures followed AVMA and FELASA guidelines for the humane treatment and euthanasia of amphibians.

3. RESULTS

3.1 Repeatability of Behavioural Traits

Short-term analysis of *Polypedates cruciger* larvae showed significant repeatabilities across most measured putative personality traits; including boldness (except for RZ), and exploration; but excluding activity, sociability and aggression (Table 3.1). Track Length (TL), RZ, CV and AT variables showed repeatability values as zero or near zero and were not statistically significant (Table 3.1). In long-term analysis of *P. cruciger* larvae; significant repeatabilities were maintained for putative traits boldness (except for RZ), exploration, sociability, and aggression, while activity (TL) and RZ representing boldness remained non-significant (Table 3.1). Among the variables, LA (in both short-term and long-term analysis) and CV and CM in long-term analysis showed particularly high and significant repeatability. Track length (TL) and RZ continued to be non-significant with zero repeatability (Table 3.1).

Several behaviors became substantially more repeatable over longer intervals. Boldness measured as emergence time (ET) decreased from moderate short-term repeatability to a lower long-term value, while avoidance behavior increased markedly, becoming highly repeatable long-term. Exploration traits followed a similar pattern: LA, CM, and general exploration (Expo) all shifted from low-to-moderate short-term repeatability to high long-term consistency. Likewise, sociability (CV) increased from low to high repeatability over time. Aggression showed weak short-term repeatability but moderate long-term consistency. Overall, *P. cruciger* exhibited stronger long-term stabilization of behavioral traits (Table 3.1).

Table 3.1 Repeatability estimates (with 95% credible intervals in parentheses) for each personality trait in common hourglass frog (*Polypedates cruciger*) measured across short-term

Personality trait	Behavioral variable	Short-term			Long-term		
		Repeatability	CI value	<i>p</i> value	Repeatability	CI value	<i>p</i> value
Activity	TL	0	(0,0.118)	0.5	0	(0,0.158)	1
Boldness	RZ	0	(0,0.013)	0.5	0	(0,0.179)	1
	ET	0.454	(0.284,0.590)	<0.001	0.281	(0.092,0.463)	0.001
	Avoid	0.200	(0.033,0.308)	0.005	0.652	(0.489,0.762)	<0.001
Exploration	LA	0.548	(0.404,0.669)	<0.001	0.822	(0.727,0.884)	<0.001
	TS	0.371	(0.204,0.506)	<0.001	0.295	(0.102,0.457)	<0.001
	CM	0.195	(0.048,0.328)	0.002	0.877	(0.797,0.916)	<0.001
	Expo	0.176	(0.033,0.314)	0.004	0.681	(0.535,0.787)	<0.001
Sociability	CV	0.124	(0,0.258)	<0.001	0.721	(0.590,0.816)	<0.001
Aggression	AT	0.088	(0,0.230)	0.099	0.288	(0.153,0.450)	<0.001

Note: Bold text indicates a significant effect ($p < .05$ in a permutation of residuals test and 95% cis not including zero). Behavioral variables: TL – track length; RZ – time spent in risky zone; ET – emergence time; LA – latency to approach an object; TS – time spent near object; CV – time spent near mirror; AT – number of attacks towards mirror; CM – latency to approach mirror; Avoid – number of avoidances done away from the mirror; Expo – number of movements done parallel to the mirror.

Short-term analysis of *Duttaphrynus melanostictus* larvae exhibited exceptionally high and significant repeatability values. Specifically, CV (sociability) and LA (exploration) repeatability approaching perfect repeatability (Table 3.2). Emergence Time; ET (boldness), also demonstrated significant repeatability. The behavioral traits such as TS, CM and Avoid too showed significant repeatabilities while TL (activity) and RZ (boldness) showed zero repeatability and were non-significant (Table 3.2).

The long-term analysis of repeatability in *D. melanostictus* larvae indicated that most personality traits were significant. In the long-term analysis of *D. melanostictus* AT (aggression) showed the highest repeatability (Table 3.2). Significant repeatabilities were showed for all the others including ET, Avoid (Boldness), LA, TS, CM, Expo (exploration), CV (sociability) and AT (aggression) except TL (activity) and RZ (boldness).

Duttaphrynus melanostictus displayed very high short-term repeatability in several traits, particularly boldness (ET), exploration (LA), and sociability (CV), all of which were close to or above 0.88. However, repeatability in these traits declined substantially over the long term,

although they generally remained significant. Exploration measures such as CM and Expo, as well as avoidance behavior, increased modestly from short- to long-term, while aggression shifted from low short-term repeatability to high long-term consistency. Activity and boldness measured by RZ remained non-repeatable in both periods. When comparing species, *D. melanostictus* showed greater short-term consistency (table 3.2), while *P. cruciger* demonstrated higher long-term repeatability across a wider set of behaviors.

Table 3.2 Repeatability estimates (with 95% credible intervals in parentheses) for each personality trait in common toad (*Duttaphrynus melanostictus*) measured across short-term (n = 50) and long-term time periods (n = 50)

Personality trait	Behavioral variable	Short-term			Long-term		
		Repeatability	CI value	<i>p</i> value	Repeatability	CI value	<i>p</i> value
Activity	TL	0	0,0.124	1	0.027	0,0.373	0.0003
Boldness	RZ	0	0,0.118	0.5	0	0,0.125	0.5
	ET	0.884	0.819,0.921	<0.001	0.379	0.225,0.520	<0.001
	Avoid	0.170	0.031,0.313	<0.001	0.431	0.259,0.560	<0.001
Exploration	LA	0.981	0.969,0.988	<0.001	0.509	0.354,0.627	<0.001
	TS	0.382	0.215,0.516	<0.001	0.176	0.041,0.322	0.004
	CM	0.195	0.049,0.337	0.002	0.473	0.311,0.601	<0.001
	Expo	0.059	0,0.194	0.192	0.488	0.325,0.621	<0.001
Sociability	CV	0.996	0.994,0.998	<0.001	0.469	0.306,0.596	<0.001
Aggression	AT	0.093	0,0.214	0.089	0.654	0.513,0.748	<0.001

Note: Bold text indicates a significant effect ($p < .05$ in a permutation of residuals test and 95% cis not including zero). Behavioral variables: TL – track length; RZ – time spent in risky zone; ET – emergence time; LA – latency to approach an object; TS – time spent near object; CV – time spent near mirror; AT – number of attacks towards mirror; CM – latency to approach mirror; Avoid – number of avoidances done away from the mirror; Expo – number of movements done parallel to the mirror.

Short-term repeatability differed markedly between *P. cruciger* and *D. melanostictus*. *Duttaphrynus melanostictus* showed consistently higher short-term repeatability across several behaviours, most notably boldness (ET), exploration (LA), and sociability (CV), all of which exhibited very high consistency (≥ 0.88). In contrast, *P. cruciger* showed generally low short-term repeatability, with only moderate values observed for ET, LA, TS, and CM. Activity (TL) and boldness (RZ) were non-repeatable in both species. Overall, *D. melanostictus* displayed much stronger immediate behavioural consistency, indicating that its personality traits are more stable over short temporal scales compared to *P. cruciger* (See Table 3.1 and 3.2).

Long-term patterns showed the opposite trend. *Polypedates cruciger* demonstrated substantially higher long-term repeatability across multiple behavioural domains, with exploration traits (LA, CM, Expo) and sociability (CV) showing strong to very strong consistency over time. In contrast, long-term repeatability in *D. melanostictus* was generally moderate and often lower than its short-term values, except for aggression (AT), which remained high. Although both species showed low long-term repeatability in activity (TL) and boldness (RZ), *P. cruciger* maintained higher long-term stability across a wider range of behaviours. These results indicate that while *D. melanostictus* exhibited greater short-term consistency, *P. cruciger* showed more stable personality expression over longer periods (See Table 3.1 and 3.2).

3.2 Correlations of Behavioural Traits

The correlation analysis for the two amphibian larval species, *P. cruciger* and *D. melanostictus*, demonstrated varying degrees and types of significant relationships between behaviors across different time scales. In the short-term analysis of *P. cruciger*, highly significant correlations were readily apparent (Table 3.3; Fig 3.1a). Emergence time (ET) also showed a positive within individual correlation with exploratory behaviour (Expo), suggesting that faster-emerging individuals explored more. A positive association was also seen between ET and aggression (AT) at both phenotypic and within-individual levels, showing that bolder individuals tended to be more aggressive. Activity (TL) showed positive phenotypic and within-individual correlations with latency to approach (LA), as well as with time spent in the risky zone (TS) and exploratory behaviour (Expo), suggesting that more active larvae were also more exploratory (Table 3.3; Fig 3.1a). Sociability (CV) also showed positive phenotypic and within-individual correlations with exploratory behaviour (Expo). Latency to approach (LA) showed positive phenotypic, within-individual, and among-individual correlations with time in the risky zone (TS), indicating a consistent behavioural link where individuals that explored more also took more risks. Finally, LA exhibited a positive phenotypic correlation with exploratory behaviour (Expo), showing that different types of exploration were positively linked. (Table 3.3; Figure 3.1a).

In the long-term analysis, sociability (CV) showed a positive within-individual correlation with emergence time (ET), suggesting that within individuals, changes in sociability were aligned with changes in boldness. A positive within-individual correlation was also observed between

ET and latency to approach (LA). Activity (TL) displayed a positive phenotypic and negative within-individual correlation with sociability (CV), indicating that more active individuals tended to be less social (Table 3.3; Fig 3.1b). A positive phenotypic and within-individual correlation was found between boldness measured by risky-zone use (RZ) and sociability (CV). Exploration (LA) showed positive correlations across phenotypic, within-individual, and among-individual levels with time spent in the risky zone (TS), and exploratory behaviour (Expo), showing broad consistency in exploration-related traits (Table 3.3; Fig 3.1b)

Table 3.3 Within individual, among individual and overall phenotypic correlations (with 95% credible intervals in parentheses) between behavioral variable in common toad (*Polypedates cruciger*) measured across short-term (n = 50) and long-term time periods (n=50)

Putative personality trait for behavioral variable 1	Behavioral variable 1	Putative personality trait for behavioral variable 2	Behavioral variable 2	Short-term correlations			Long-term correlations		
				Phenotypic	Within individual	Among individual	Phenotypic	Within individual	Among individual
Boldness	ET	Activity	TL	-0.082 (-0.209,0.062)	-0.199 (-0.350,-0.049)	0.987 (-0.997,0.999)	-0.001 (-0.167,0.154)	-0.014 (-0.193,0.159)	0.859 (-0.999,0.999)
		Boldness	RZ	-0.058 (-0.198,0.085)	-0.046 (-0.204,0.115)	-0.705 (-0.997,0.972)	0.957 (-0.942,0.999)	0.021 (-0.159,0.201)	0.900 (-0.962,0.999)
			Avoid	-0.051 (-0.218,0.115)	-0.037 (-0.191,0.117)	-0.068 (-0.372,0.253)	-0.007 (-0.151,0.151)	-0.051 (-0.262,0.156)	0.424 (-0.427,0.559)
		Sociability	CV	-0.052 (-0.99,0.59)	-0.075 (-0.19,0.15)	0.341 (-0.95,0.99)	-0.120 (-0.269,0.040)	-0.359 (-0.534,-0.167)	0.996 (-0.997,0.999)
		Exploration	LA	-0.07 (-0.25,0.11)	-0.14 (-0.29,0.02)	0.51 (-0.56,0.55)	-0.026 (-0.185,0.139)	0.248 (0.026,0.445)	-0.996 (-0.999,0.998)
		Exploration	TS	0.071 (0.017,0.081)	-0.008 (-0.17,0.16)	0.011 (-0.38,0.43)	0.023 (-0.133,0.186)	0.049 (-0.164,0.254)	-0.824 (-0.998,0.995)
			Expo	-0.049 (-0.213,0.113)	0.103 (0.058,0.250)	0.046 (-0.309,0.416)	0.023 (-0.133,0.186)	0.049 (-0.164,0.254)	-0.824 (-0.998,0.995)
			CM	0.476 (0.368,-0.579)	0.479 (-0.371,0.582)	0.130 (-0.976,0.998)	0.066 (-0.070,0.195)	0.118 (-0.090,0.310)	0.002 (-0.409,0.436)
		Aggression	AT	0.476 (0.368,0.579)	0.479 (0.371,0.582)	0.130 (-0.976,0.998)	-0.026 (-0.185,0.139)	0.248 (-0.026,0.445)	-0.996 (-0.999,0.998)
Activity	TL	Boldness	RZ	-0.999 (-0.999,0.947)	0.085 (-0.088,0.014)	-0.014 (-0.153,0.115)	0.105 (-0.053,0.266)	-0.107 (-0.155,0.269)	-0.160 (-0.999,0.981)
			Avoid	0.108 (-0.028,0.242)	0.126 (-0.020,0.270)	-0.949 (-0.999,0.990)	-0.007 (-0.142,0.159)	-0.052 (-0.259,0.149)	0.094 (-0.428,0.543)
		Sociability	CV	-0.024 (-0.162,0.101)	0.006 (-0.141,0.155)	-0.991 (-0.999,0.995)	0.124 (0.039,0.272)	-0.357 (-0.532,-0.152)	0.995 (-0.994,0.999)
		Exploration	LA	0.243 (0.118,0.369)	0.279 (0.134,0.419)	0.142 (-0.367,0.580)	0.129 (0.064,0.192)	0.036 (-0.141,0.238)	0.995 (-0.998,0.999)
			TS	0.156 (0.018,0.257)	0.223 (0.078,0.360)	0.000 (-0.976,0.998)	0.024 (-0.137,0.182)	0.564 (-0.156,0.265)	-0.909 (-0.999,0.994)
			Expo	0.476 (0.368,0.579)	0.479 (0.371,0.582)	0.130 (-0.976,0.998)	0.066 (-0.070,0.195)	0.118 (-0.090,0.310)	0.002 (-0.409,0.436)

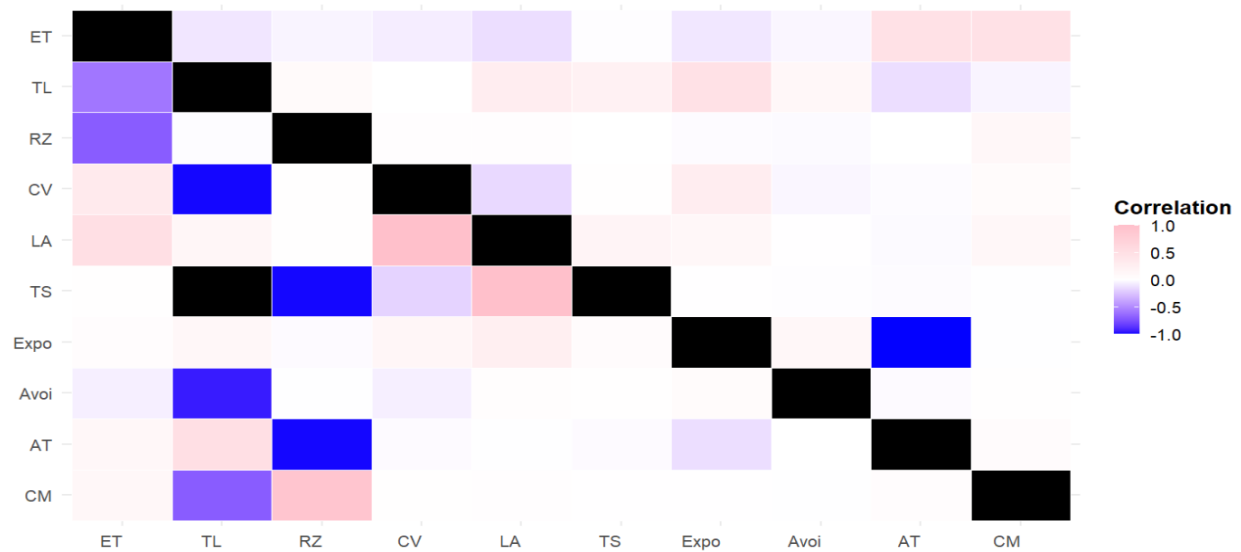
Table 3.3 (continued)

			CM	-0.058 (-0.198,0.085)	-0.046 (-0.204,0.115)	-0.705 (-0.997,0.972)	0.016 (-0.091,0.127)	0.026 (-0.155,0.209)	0.023 (-0.469,0.544)
		Aggression	AT	-0.07 (-0.25,0.11)	-0.14 (-0.29,0.02)	0.51 (-0.36,0.35)	0.024 (-0.137,0.182)	0.564 (-0.156,0.265)	-0.909 (-0.960,0.954)
Boldness	RZ	Boldness	Avoid	-0.016 (-0.127,0.092)	-0.023 (-0.179,0.126)	-0.005 (-0.338,0.316)	0.016 (-0.091,0.127)	0.026 (-0.155,0.209)	0.023 (-0.469,0.544)
		Sociability	CV	0.016 (-0.091,0.127)	0.026 (-0.155,0.209)	0.023 (-0.469,0.544)	0.184 (0.027,0.341)	0.184 (0.010,0.372)	0.947 (-0.796,0.999)
		Exploration	LA	0.016 (-0.091,0.127)	0.026 (-0.135,0.209)	0.023 (-0.466,0.544)	0.104 (-0.063,0.260)	0.125 (-0.059,0.314)	0.928 (-0.860,0.999)
			TS	-0.024 (-0.162,0.101)	0.006 (-0.141,0.155)	-0.991 (-0.999,0.997)	0.019 (-0.145,0.178)	-0.036 (-0.211,0.157)	0.917 (-0.923,0.999)
			Expo	-0.013 (-0.138,0.111)	-0.015 (-0.171,0.131)	-0.019 (-0.495,0.443)	0.041 (-0.068,0.149)	0.070 (-0.117,0.247)	0.015 (-0.428,0.457)
			CM	0.104 (-0.063,0.260)	0.125 (-0.059,0.314)	0.928 (-0.860,0.999)	0.041 (-0.068,0.149)	0.070 (-0.117,0.247)	0.015 (-0.428,0.457)
		Aggression	AT	-0.024 (-0.162,0.101)	0.006 (-0.141,0.155)	-0.991 (-0.999,0.997)	-0.007 (-0.142,0.159)	-0.052 (-0.259,0.149)	0.094 (-0.428,0.543)
Sociability	CV	Boldness	Avoid	-0.050 (-0.211,0.113)	-0.037 (-0.181,0.129)	-0.069 (-0.365,0.251)	0.309 (0.104,0.499)	0.205 (0.009,0.380)	0.361 (0.075,0.618)
		Exploration	LA	0.053 (-0.09,0.204)	-0.163 (-0.315,0.225)	0.993 (-0.967,0.999)	0.344 (0.121,0.549)	0.032 (-0.169,0.223)	0.412 (0.152,0.649)
			TS	0.015 (-0.124,0.152)	0.023 (-0.146,0.189)	-0.186 (-0.998,0.993)	0.202 (0.026,0.377)	0.027 (-0.128,0.258)	0.551 (0.121,0.998)
			Expo	0.243 (0.118,0.369)	0.279 (0.134,0.419)	0.142 (-0.367,0.580)	0.269 (0.065,0.466)	0.073 (-0.128,0.258)	0.361 (0.083,0.624)
			CM	0.041 (-0.068,0.149)	0.070 (-0.117,0.247)	0.015 (-0.428,0.457)	0.142 (-0.367,0.580)	0.269 (-0.065,0.466)	0.073 (-0.128,0.258)
		Aggression	AT	-0.013 (-0.138,0.111)	-0.015 (-0.171,0.131)	-0.019 (-0.495,0.443)	0.184 (0.027,0.341)	0.184 (0.010,0.372)	0.947 (-0.796,0.999)
Exploration	LA	Boldness	Avoid	0.016 (-0.161,0.199)	-0.003 (-0.161,0.199)	0.033 (-0.277,0.342)	-0.019 (-0.270,0.220)	-0.029 (-0.227,0.159)	-0.020 (-0.329,0.269)
		Exploration	TS	0.511 (0.388,0.634)	0.174 (0.125,0.321)	0.999 (0.959,0.9992)	0.387 (0.228,0.526)	0.423 (0.237,0.579)	0.895 (0.481,0.999)
			Expo	0.177 (0.012,0.344)	0.124 (-0.027,0.274)	0.256 (-0.088,0.574)	0.258 (0.026,0.477)	0.005 (-0.189,0.198)	0.318 (0.023,0.561)

Table 3.3 (continued)

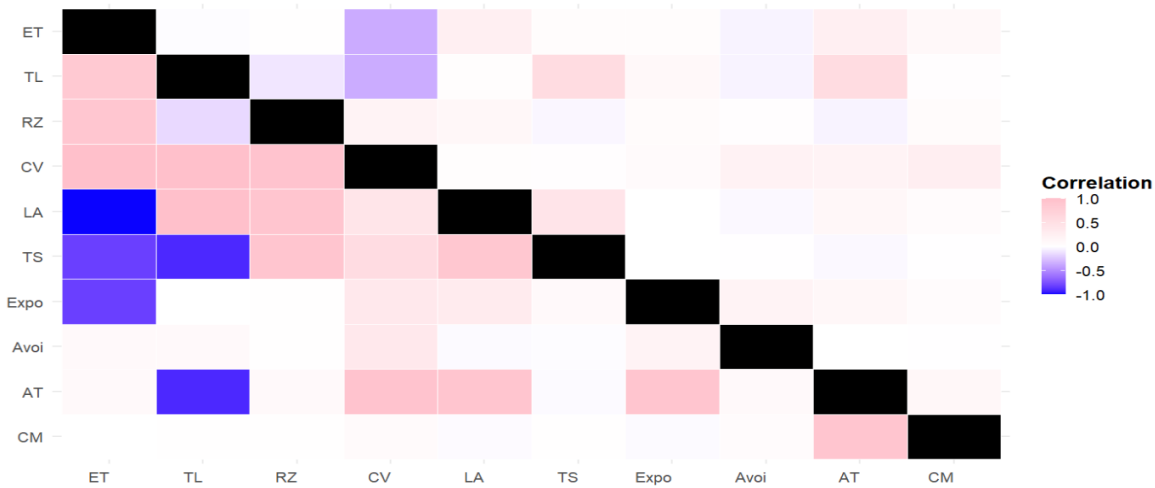
			CM	0.177 (-0.011 ,0.344)	0.124 (-0.027,0.274)	0.028 (-0.123,0.195)	0.009 (-0.140,0.164)	0.065 (-0.303,0.435)	-0.019 (-0.270,0.220)
		Aggression	AT	-0.016 (-0.127,0.092)	-0.023 (-0.179,0.126)	-0.005 (-0.338,0.316)	0.104 (-0.063,0.260)	0.125 (-0.059,0.314)	0.928 (-0.860,0.999)
Exploration	TS	Boldness	Avoid	-0.002 (-0.174,0.162)	-0.007 (-0.169,0.156)	0.013 (-0.382,0.420)	-0.005 (-0.173,0.175)	-0.003 (-0.206,0.189)	-0.012 (-0.382,0.367)
		Exploration	Expo	0.028 (-0.123,0.195)	0.009 (-0.140,0.164)	0.065 (-0.303,0.435)	0.044 (-0.123,0.215)	0.007 (-0.179,0.126)	0.099 (-0.262,0.441)
			CM	0.013 (-0.382,0.420)	-0.005 (-0.173,0.175)	-0.003 (-0.206,0.189)	-0.012 (-0.382,0.367)	0.009 (-0.140,0.164)	0.012 (-0.382,0.367)
		Aggression	AT	-0.013 (-0.138,0.111)	-0.015 (-0.171,0.131)	-0.019 (-0.495,0.443)	-0.019 (-0.270,0.220)	-0.029 (-0.227,0.159)	-0.020 (-0.329,0.269)
Exploration	Expo	Boldness	Avoid	0.101 (-0.056,0.251)	0.126 (-0.027,0.270)	0.068 (-0.245,0.359)	0.192 (-0.024,0.392)	0.193 (0.056,0.371)	0.191 (-0.122,0.470)
		Aggression	AT	-0.050 (-0.211,0.113)	-0.991 (-0.999,0.958)	-0.14 (-0.29,0.302)	0.104 (-0.263,0.260)	0.125 (-0.259,0.314)	0.928 (-0.960,0.999)
		Exploration	CM	0.013 (-0.382,0.420)	-0.005 (-0.173,0.175)	-0.003 (-0.206,0.189)	0.009 (-0.140,0.164)	0.065 (-0.303,0.435)	-0.023 (-0.179,0.126)
Boldness	Avoid	Aggression	AT	-0.07 (-0.25,0.56)	-0.019 (-0.495,0.443)	0.006 (-0.141,0.155)	0.044 (-0.123,0.215)	0.007 (-0.063,0.260)	0.099 (-0.262,0.441)
		Exploration	CM	-0.007 (-0.169,0.156)	0.013 (-0.382,0.420)	-0.005 (-0.173,0.175)	0.028 (-0.123,0.195)	0.009 (-0.140,0.164)	0.065 (-0.303,0.435)
Aggression	AT	Exploration	CM	0.009 (-0.140,0.164)	0.065 (-0.303,0.435)	0.044 (-0.123,0.215)	0.104 (-0.163,0.260)	0.125 (-0.259,0.314)	0.928 (-0.960,0.999)

Note: Bold text indicates a significant effect (95% cis not including zero). Behavioral variables: TL – track length; RZ – time spent in risky zone; ET – emergence time; LA – latency to approach an object; TS – time spent near object; CV – time spent near mirror; AT – number of attacks towards mirror; CM – latency to approach mirror; Avoid – number of avoidances done away from the mirror; Expo – number of movements done parallel to the mirror.



(a)

Figure 3.1 a) Within-individual (above diagonal) and among-individual (below diagonal) correlations between behavioural variables in *Polypedates cruciger* short-term time period



(b)

Figure 3.1 b) Within-individual (above diagonal) and among-individual (below diagonal) correlations between behavioural variables in *Polypedates cruciger* long-term time period

Note: Behavioral variables: TL – track length; RZ – time spent in risky zone; ET – emergence time; LA – latency to approach an object ; TS – time spent near object; CV – time spent near mirror; AT – number of attacks towards mirror; CM – latency to approach mirror; Avoi – number of avoidances done away from the mirror; Expo – number of movements done parallel to the mirror.

The short-term correlations in *D. melanostictus* indicated strong linkages between activity, boldness, and exploration. In short-term measurements, emergence time (ET) and (LA) showed positive within-individual and among-individual correlations, indicating that faster-emerging individuals were also more explorative across both scales. Activity (TL) showed a positive phenotypic and within-individual correlation with time spent in the risky-zone (RZ). Latency to approach (LA) showed a positive within-individual correlation with activity (TL). Exploratory behaviour (Expo) displayed a positive within-individual correlation with sociability (CV). Latency to approach (LA) also showed positive among-individual correlations with time spent in the risky zone (TS) (Table 3.4; Fig 3.2a). Exploration (Expo) showed a positive phenotypic correlation with avoidance behaviour. A positive phenotypic, within-individual, and among-individual correlation appeared between exploratory behaviour (Expo) and aggression (AT) (Table 3.4; Figure 3.2a).

The long-term analysis for *D. melanostictus* continued to show complex correlation structures. In the long term, activity (TL) showed a positive phenotypic and within-individual correlation with risky-zone use (RZ), indicating that more active larvae also tended to display higher boldness. Time spent in the risky zone (RZ) also showed a positive among-individual correlation with exploratory behaviour (Expo) while displaying a positive phenotypic and within-individual correlation with sociability (CV). Sociability (CV) showed a positive phenotypic, within-individual, and among-individual correlation with LA and phenotypic and within-individual correlation with aggression (AT) (Table 3.4; Figure 3.2b) suggesting consistent behavioural linkages across response types. Latency to approach (LA) also displayed positive phenotypic and within-individual correlations with exploratory behavior and phenotypic, within individual and among individual correlations with TS. Activity (TL) showed a positive among-individual correlation with emergence time (ET), indicating a long-term behavioural syndrome linking activity and boldness. Latency to approach (LA) showed a positive phenotypic and within individual correlations with aggression (AT). (Table 3.4; Figure 3.2b). The two amphibian species were showing similar trends in within individual and among individual correlations but there were major differences in the strengths of these correlations (see Figure 3.1 and Figure 3.2).

Table 3.4 Within individual, among individual and overall phenotypic correlations (with 95% credible intervals in parentheses) between behavioral variable in common toad (*Duttaphrynus melanostictus*) measured across short-term (n = 50) and long-term time periods (n=50)

Putative personality trait for behavioral variable 1	Behavioral variable 1	Putative personality trait for behavioral variable 2	Behavioral variable 2	Short-term correlations			Long -term Correlations		
				Phenotypic	Within individual	Among individual	Phenotypic	Within individual	Among individual
Boldness	ET	Activity	TL	0.066 (-0.060,0.207)	-0.018 (-0.182,0.142)	0.996 (-0.989,0.999)	0.058 (-0.098,0.213)	-0.193 (- 0.341,0.029)	0.099 (0.095,0.999)
		Boldness	RZ	0.033 (-0.116,0.176)	0.049 (-0.009,0.212)	0.312 (-0.309,0.993)	0.004 (-0.136,0.150)	-0.084 (-0.229,0.073)	0.936 (-0.504,0.999)
		Avoid		-0.001 (-0.261,0.141)	-0.065 (-0.221,0.091)	-0.076 (-0.385,0.269)	0.016 (-0.146,0.189)	-0.091 (-0.247,0.062)	0.145 (-0.188,0.459)
		Sociability	CV	-0.181 (-0.419,0.093)	0.006 (-0.159,0.161)	-0.194 (-0.448,0.094)	-0.096 (-0.261,0.079)	-0.219 (-0.373,0.064)	0.092 (-0.314,0.477)
		Exploration	LA	-0.001 (-0.264,0.261)	0.227 (0.163,0.584)	0.002 (0.001,0.276)	-0.046 (-0.221,0.125)	-0.062 (-0.219,0.099)	-0.024 (-0.388,0.361)
			TS	-0.006 (-0.214,0.185)	0.001 (-0.163,0.163)	-0.012 (-0.346,0.331)	-0.037 (-0.186,0.108)	-0.028 (-0.202,0.149)	-0.117 (-0.997,0.857)
			Expo	-0.072 (-0.278,0.132)	0.009 (-0.147,0.164)	-0.125 (-0.434,0.203)	0.024 (-0.156,0.184)	-0.044 (-0.194,0.118)	0.101 (-0.224,0.415)
			CM	0.104 (-0.063,0.260)	0.125 (-0.059,0.314)	0.928 (-0.860,0.999)	0.002 (-0.175,0.176)	0.048 (-0.109,0.215)	-0.069 (-0.449,0.316)
		Aggression	AT	-0.019 (-0.216,0.169)	0.024 (-0.135,0.184)	-0.055 (-0.381,0.279)	-0.003 (-0.179,0.184)	-0.137 (-0.296,0.025)	0.129 (-0.217,0.481)
Activity	TL	Boldness	Avoid	-0.019 (-0.133,0.100)	-0.017 (-0.261,0.046)	0.264 (-0.092,0.590)	0.021 (-0.133,0.164)	0.098 (-0.055,0.256)	-0.124 (-0.479,0.247)
			RZ	0.052 (0.462,0.653)	0.566 (0.463,0.654)	0.084 (-0.957,0.995)	0.314 (0.178,0.433)	0.373 (0.238,0.500)	-0.565 (-0.998,0.961)
		Sociability	CV	0.073 (-0.064,0.202)	-0.001 (-0.163,0.151)	0.996 (-0.988,0.999)	0.000 (-0.142,0.149)	0.119 (-0.163,0.291)	-0.992 (-0.999,0.468)
		Exploration	LA	-0.034 (-0.164,0.105)	0.167 (0.177,0.319)	-0.994 (-0.999,0.996)	0.033 (-0.106,0.175)	0.014 (-0.177,0.201)	0.930 (-0.999,0.999)
			TS	-0.045 (-0.199,0.104)	0.001 (-0.151,0.170)	-0.205 (-0.709,0.326)	0.079 (-0.062,0.219)	0.096 (-0.098,0.268)	-0.114 (-0.997,0.982)
			Expo	-0.035 (-0.147,0.083)	-0.046 (-0.194,0.108)	0.005 (-0.384,0.384)	0.032 (-0.119,0.166)	0.078 (-0.083,0.234)	-0.044 (-0.376,0.317)
			CM	-0.017 (-0.261,0.046)	0.264 (-0.092,0.590)	0.021 (-0.133,0.164)	0.098 (-0.055,0.256)	0.119 (-0.163,0.291)	0.996 (-0.988,0.999)
		Aggression	AT	0.55 (-0.088,0.202)	0.089 (-0.033,0.260)	0.016 (-0.847,0.926)	-0.994 (-0.999,0.996)	0.033 (-0.106,0.175)	0.014 (-0.177,0.201)
Boldness	RZ	Boldness	Avoid	-0.124 (-0.391,0.154)	0.014 (-0.150,0.171)	-0.123 (-0.396,0.156)	0.036 (-0.069,0.141)	0.098 (-0.155,0.256)	-0.124 (-0.479,0.247)

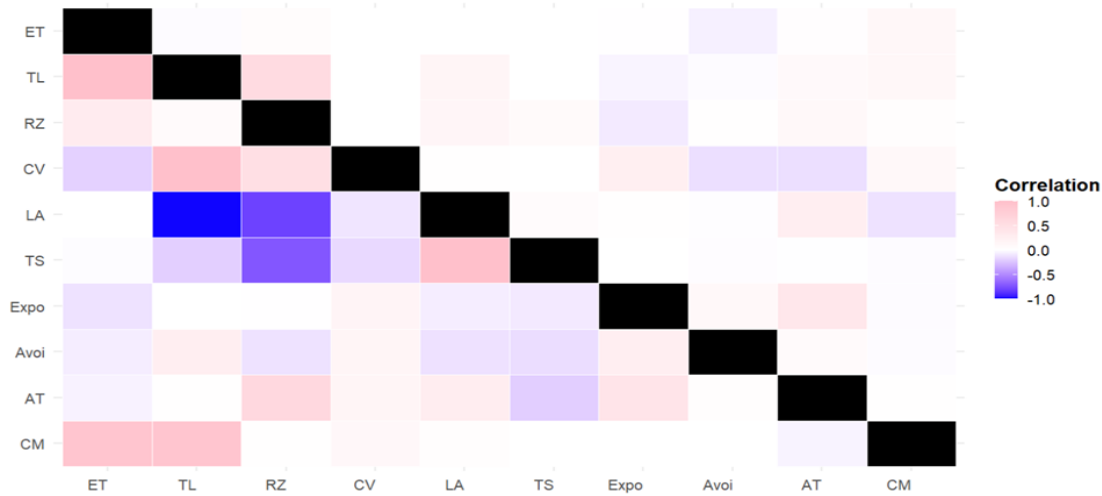
Table 3.4 (continued)

		Exploration	CM	0.104 (-0.063,0.260)	0.125 (-0.059,0.314)	0.928 (-0.860,0.999)	0.015 (-0.149,0.122)	0.023 (-0.195,0.162)	0.000 (-0.999,0.998)
			LA	-0.044 (-0.194,0.095)	0.153 (-0.006,0.304)	-0.812 (-0.999,0.794)	0.073 (-0.064,0.214)	0.018 (-0.050,0.252)	0.111 (-0.972,0.992)
			TS	0.038 (-0.096,0.178)	0.087 (-0.064,0.241)	-0.729 (-0.997,0.911)	0.021 (-0.117,0.158)	0.028 (-0.116,0.179)	-0.238 (-0.988,0.964)
			Expo	-0.066 (-0.178,0.055)	-0.088 (-0.231,0.069)	-0.003 (-0.380,0.386)	0.078 (-0.018,0.179)	0.122 (-0.133,0.263)	0.029 (0.004,0.404)
		Sociability	CV	0.029 (-0.114,0.175)	0.006 (-0.150,0.165)	0.514 (-0.931,0.997)	0.232 (0.310,0.355)	0.201 (0.049,0.342)	0.945 (-0.213,0.999)
		Aggression	AT	0.110 (-0.004,0.241)	0.119 (-0.037,0.271)	0.618 (-0.587,0.975)	0.111 (-0.013,0.229)	0.120 (-0.134,0.271)	0.630 (-0.588,0.970)
Sociability	CV	Boldness	Avoid	0.097 (-0.112,0.305)	-0.137 (-0.291,0.017)	0.165 (-0.158,0.480)	0.319 (0.158,0.479)	0.348 (0.208,0.487)	0.297 (-0.016,0.579)
		Exploration	LA	-0.121 (-0.392,0.150)	0.018 (-0.146,0.174)	-0.112 (-0.394,0.154)	0.337 (0.175,0.495)	0.289 (0.134,0.433)	0.398 (0.059,0.684)
			TS	-0.095 (-0.298,0.114)	0.003 (-0.162,0.159)	-0.161 (-0.486,0.167)	0.203 (0.058,0.337)	0.129 (-0.137,0.298)	0.709 (-0.003,0.998)
			Expo	0.120 (-0.087,0.324)	0.262 (0.116,0.405)	0.168 (-0.144,0.469)	0.272 (0.105,0.434)	0.265 (0.119,0.411)	0.287 (-0.026,0.556)
			CM	0.618 (-0.587,0.975)	0.111 (-0.013,0.229)	0.120 (-0.034,0.271)	0.014 (-0.150,0.171)	0.123 (-0.396,0.156)	0.036 (-0.069,0.141)
		Aggression	AT	0.097 (-0.112,0.305)	-0.137 (-0.291,0.017)	0.165 (-0.158,0.480)	0.232 (0.051,0.409)	0.217 (0.063,0.371)	0.258 (-0.075,0.566)
Exploration	LA	Boldness	Avoid	-0.081 (-0.291,0.122)	-0.009 (-0.163,0.151)	-0.127 (-0.447,0.186)	-0.025 (-0.205,0.160)	0.032 (-0.127,0.189)	-0.078 (-0.405,0.234)
		Exploration	TS	0.614 (0.503,0.725)	0.056 (-0.103,0.213)	0.995 (0.963,0.999)	0.424 (0.299,0.539)	0.348 (0.206,0.499)	0.938 (0.360,0.999)
			Expo	-0.047 (-0.252,0.172)	0.018 (-0.137,0.173)	-0.078 (-0.403,0.239)	0.206 (0.015,0.371)	0.206 (0.057,0.356)	0.205 (-0.093,0.501)
			CM	0.014 (-0.150,0.171)	-0.123 (-0.396,0.156)	0.036 (-0.069,0.141)	0.027 (-0.161,0.211)	0.050 (-0.108,0.214)	0.001 (-0.361,0.350)
		Aggression	AT	0.278 (0.098,0.460)	0.272 (0.120,0.415)	0.292 (-0.030,0.579)	0.280 (0.090,0.480)	0.271 (0.111,0.408)	0.295 (-0.023,0.581)
Exploration	TS	Boldness	Avoid	-0.063 (-0.224,0.102)	-0.011 (-0.164,0.148)	-0.146 (-0.475,0.209)	-0.018 (-0.156,0.109)	0.002 (-0.154,0.161)	-0.085 (-0.461,0.314)
		Exploration	Expo	-0.036 (-0.193,0.133)	-0.001 (-0.162,0.150)	-0.093 (-0.419,0.245)	-0.047 (-0.252,0.172)	0.018 (-0.137,0.173)	-0.078 (-0.403,0.239)
			CM	-0.146 (-0.475,0.209)	-0.018 (-0.156,0.109)	0.002 (-0.154,0.161)	0.004 (-0.134,0.159)	0.024 (-0.150,0.190)	-0.094 (-0.991,0.808)
		Aggression	AT	-0.037 (-0.185,0.106)	0.004 (-0.167,0.165)	-0.211 (-0.929,0.443)	-0.036 (-0.188,0.106)	0.004 (-0.154,0.175)	-0.203 (-0.938,0.437)
Exploration	Expo	Boldness	Avoid	0.174 (0.012,0.333)	0.113 (-0.038,0.261)	0.263 (-0.044,0.549)	0.060 (-0.072,0.191)	0.072 (-0.077,0.238)	0.053 (-0.321,0.398)

Table 3.4 (continued)

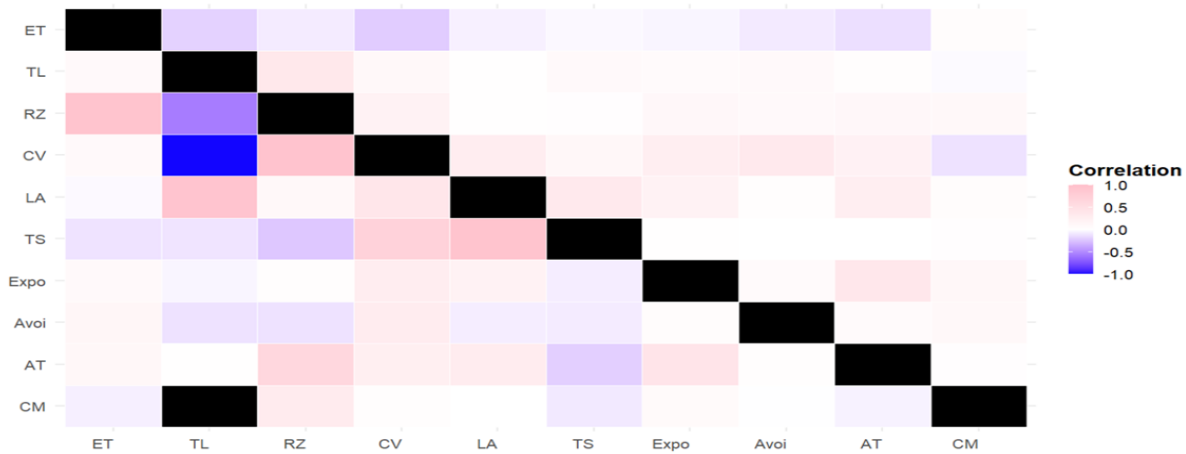
		Aggression	AT	0.406 (0.226,0.555)	0.377 (0.236,0.507)	0.427 (0.165,0.659)	0.406 (0.231,0.565)	0.379 (0.239,0.505)	0.424 (0.159,0.659)
		Exploration	CM	-0.146 (-0.475,0.209)	-0.018 (-0.156,0.109)	0.002 (-0.154,0.161)	0.104 (-0.081,0.277)	0.126 (-0.129,0.279)	0.086 (-0.228,0.389)
Boldness	Avoid	Aggression	AT	0.051 (-0.142,0.249)	0.074 (-0.082,0.232)	0.037 (-0.266,0.353)	0.053 (-0.146,0.247)	0.074 (-0.079,0.235)	0.039 (-0.274,0.344)
		Exploration	CM	-0.146 (-0.475,0.209)	-0.018 (-0.156,0.109)	0.002 (-0.154,0.161)	0.052 (-0.121,0.239)	0.111 (-0.052,0.261)	-0.004 (-0.322,0.329)
Aggression	AT	Exploration	CM	-0.0175 (-0.219,0.165)	0.023 (-0.136,0.183)	-0.052 (-0.380,0.279)	-0.021 (-0.212,0.182)	0.024 (-0.135,0.183)	-0.057 (-0.387,0.279)

Note: Bold text indicates a significant effect (95% cis not including zero). Behavioral variables: TL – track length; RZ – time spent in risky zone; ET – emergence time; LA – latency to approach an object ; TS – time spent near object; CV – time spent near mirror; AT – number of attacks towards mirror; CM – latency to approach mirror; Avoid – number of avoidances done away from the mirror; Expo – number of movements done parallel to the mirror.



(a)

Figure 3.2 a) Within-individual (above diagonal) and among-individual (below diagonal) correlations between behavioural variables in *Duttaphrynus melanostictus* short-term time period



(b)

Figure 3.2 b) Within-individual (above diagonal) and among-individual (below diagonal) correlations between behavioural variables in *Duttaphrynus melanostictus* long-term time period

Note: Behavioral variables: TL – track length; RZ – time spent in risky zone; ET – emergence time; LA – latency to approach an object ; TS – time spent near object; CV – time spent near mirror; AT – number of attacks towards mirror; CM – latency to approach mirror; Avoi – number of avoidances done away from the mirror; Expo – number of movements done parallel to the mirror.

4. DISCUSSION

Overall, this study provides the first comprehensive assessment of personality traits and behavioural syndrome structure in the larvae of *Polypedates cruciger* and *Duttaphrynus melanostictus*. By combining repeatability analyses across short and long temporal scales with detailed phenotypic, within-individual, and among-individual correlations, the study demonstrates that amphibian larvae exhibit consistent individual differences in behaviour. However, the strength, stability, and integration of these behaviours differ markedly between species. *Polypedates cruciger* showed low short-term but high long-term stability in exploration, sociability, and aggression, suggesting consolidation of behavioural traits over development. In contrast, *D. melanostictus* displayed exceptionally high repeatability in the short term - especially in exploration, sociability, and boldness - but weaker long-term stability, pointing to a behavioural strategy characterised by short-term rigidity but long-term flexibility. Across both species, activity and risky-zone use remained non-repeatable, indicating high behavioural plasticity or measurement limitations for these traits. Correlation analyses revealed complex behavioural syndromes, with proactive-like trait combinations appearing more strongly in *P. cruciger*, while *D. melanostictus* showed a more diffuse pattern of behavioural integration. These differences have ecological significance, particularly in relation to competitive dynamics, responsiveness to environmental stressors, and potential implications for conservation and management of amphibian populations.

Understanding consistent individual variation in behaviour is central to modern behavioural ecology, as such variation influences survival, competitive performance, and ecological interactions across species (Reale et al., 2007; Sih et al., 2004; Wolf and Weissing, 2012). The present study offers an in-depth evaluation of behavioural repeatability and syndrome structure in two Sri Lankan amphibian larvae, providing new insight into the emergence of personality traits during early development. Importantly, the results show that amphibian larvae are not behaviourally uniform. Instead, they express stable, individual-specific tendencies that persist over time, though to different degrees depending on species and trait type. This supports the growing recognition that personality begins early in ontogeny and can shape ecological outcomes even before metamorphosis (Stamps and Groothuis, 2010; Walls and Gabor, 2019; Pathirana et al., 2025).

In *P. cruciger*, long-term repeatability was high for exploration (LA, TS, CM, Expo), sociability (CV), and some aspects of boldness, whereas short-term values were comparatively lower. This pattern indicates that certain behavioural dispositions may strengthen or stabilise through development, perhaps reflecting increasing consistency in foraging, sensory processing, or social interactions as larvae grow. Conversely, behaviours linked to activity (TL) and risky-zone use (RZ) showed zero repeatability in both periods. The lack of repeatability in these behaviours mirrors findings in many fish and amphibian studies where movement-based or spatial-positioning metrics show high within-individual variability and low across-individual consistency (Toms et al., 2010; Conrad et al., 2011). This suggests that some behavioural measures may be more sensitive to momentary conditions or internal states, such as hunger or perceived threat, and therefore weaker proxies for personality.

Duttaphrynus melanostictus displayed the opposite temporal pattern. The species showed extremely high repeatability in the short term for exploration, sociability, and boldness, with some values approaching perfect repeatability. Such striking short-term consistency has been reported in studies where behavioural traits are closely tied to physiological state or immediate responses to stimuli (Conrad et al., 2011; Downs et al., 2014). However, long-term repeatability declined substantially across almost all traits. This suggests a behavioural strategy that is consistent over short durations but more flexible in the long term, possibly allowing individuals to adjust behavioural responses as environments shift. This flexibility could be advantageous in unpredictable habitats, where variable hydrology, fluctuating predation pressure, or shifting resource availability require behavioural adjustment (Koolhaas et al., 1999; Niemela and Dingemanse, 2018).

Taken together, these temporal patterns highlight distinct behavioural strategies between the two species. *Polypedates cruciger* follows a strategy of long-term consistency, which may enhance performance in stable lentic habitats where predictable conditions favour consistent foraging and competitive strategies. In contrast, *D. melanostictus* expresses short-term behavioural stability combined with long-term plasticity, making it potentially more adaptable in environments that fluctuate or where rapid developmental adjustments are required.

The hierarchical partitioning of covariance confirmed the existence of complex behavioral syndromes aligned with the theoretical proactive-reactive personality axis (Koprivnikar et al.,

2012; Sih et al., 2004). The consistent among-individual correlation linking Exploration and Sociability in *P. cruciger* confirms a stable, proactive strategy. This proactive profile aligns with the Pace-of-Life Syndrome (POLS) hypothesis (Godin et al., 2012; Dammhahn et al., 2018), which links consistent proactive behavior (active, bold) to faster life histories. In *Rana dalmatina*, support for POLS was found as highly active tadpoles developed faster and metamorphosed earlier can be taken as an example.

Furthermore, our models detected instances of joint plasticity (significant within-individual correlations). For example, negative within-individual correlations were found between boldness (ET) and sociability (CV), and between activity (TL) and sociability (CV) in *P. cruciger* long-term analysis. These patterns likely reflect short-term internal trade-offs in immediate risk or energetic allocation across different contexts (Downs & Dochtermann, 2014).

We also found support that behavioral precision, quantified as Intra-Individual Behavioural Variation (IIV), is relevant to life history. The finding that low Intra-Individual Behavioural Variation (IIV) in activity correlates with higher relative growth rates in predictable environments (supporting a high pace-of-life strategy, as observed in *Rana dalmatina*), provides context for the competitive dynamics between the Sri Lankan anurans. The endemic *P. cruciger*, confirmed as the superior competitor that reduces *D. melanostictus* growth, maintains a stable, proactive personality profile. This high stability suggests low IIV and supports its efficient, successful strategy for resource exploitation. In sharp contrast, the inferior competitor, *D. melanostictus*, exhibits highly transient behavioral consistency, with its exploration repeatability.

Importantly, the disparity between phenotypic and among-individual correlations highlights the limitations of inferring behavioural syndromes solely from overall phenotypic data (Brommer, 2013; Dingemanse et al., 2012). Phenotypic correlations often masked underlying individual-level patterns. For example, in the study comparing fish species, phenotypic and among-individual correlations were largely concordant in direction but differed markedly in strength; the phenotypic correlations were always much smaller in magnitude. If researchers had only examined phenotypic correlations, they likely would have made incorrect inferences about the occurrence or true strength of behavioral syndromes in these amphibian larvae. This strongly emphasizes the need for hierarchical partitioning of covariance using multivariate mixed-effects models for accurately

characterizing personality structures in amphibians (Dingemanse & Dochtermann, 2013; Pathirana et al., 2025; Pathirana et al., 2024; Brommer & Class, 2017).

Correlation analyses provided further insight into the organisation of behavioural syndromes. Both species exhibited complex networks of correlated behaviours, but the strength and structure of these networks differed. In *P. cruciger*, exploration traits (LA, TS, Expo) were consistently and strongly correlated with sociability (CV) at phenotypic, within-individual, and among-individual levels, revealing a clear proactive-like behavioural profile. This behavioural axis aligns with the proactive-reactive personality framework and the pace-of-life syndrome (POLS), where proactive individuals are typically more active, exploratory, bold, and competitive, and often linked to faster growth and high metabolic rates (Dammhahn et al., 2018; Sih et al., 2004). The presence of strong among-individual correlations suggests that these patterns reflect stable differences between individuals, rather than transient states or environmental effects.

Within-individual correlations uncovered additional layers of behavioural organisation in *P. cruciger*. Negative correlations, such as those between activity and sociability or between boldness (ET) and sociability, suggest short-term behavioural trade-offs. These likely reflect immediate constraints in risk allocation or energy expenditure, where increases in a particular behaviour necessitate compensatory decreases in another (Downs and Dochtermann, 2014). These within-individual dynamics highlight the importance of analysing multiple levels of behavioural covariance, as phenotypic correlations alone mask the underlying contributions of individual variation.

Phenotypic correlations often misrepresented the true strength or direction of behavioural syndromes in both species, reinforcing known limitations of relying on surface-level behavioural associations (Brommer, 2013; Dingemanse et al., 2012). For example, several behavioural correlations were weak or non-significant phenotypically but strong at the among-individual level. If only phenotypic correlations had been considered, researchers might have incorrectly concluded that behavioural syndromes were absent or weak. The results therefore emphasise the need for multivariate mixed-effects approaches to accurately partition sources of covariance and reveal the true architecture of behavioural syndromes (Dingemanse and Dochtermann, 2013; Brommer and Class, 2017; Pathirana et al., 2025).

The ecological relevance of these findings becomes clear when viewed in the context of species interactions and competition. Previous research shows that *P. cruciger* is a superior competitor to *D. melanostictus*, with the presence of *P. cruciger* reducing the growth and development of *D. melanostictus* larvae while benefitting its own growth when reared together (Ukuwela and Ranawana, 2011). The behavioural results of this study provide a mechanism for these ecological outcomes. The proactive personality profile of *P. cruciger*, characterised by high and stable exploration and sociability, likely enhances resource acquisition and spatial dominance. Such traits promote feeding efficiency and competitive interference, particularly in lentic habitats where food distribution can be patchy (Alford, 1999). In comparison, the behavioural flexibility of *D. melanostictus* may confer advantages in environments where conditions fluctuate, but is less effective in direct competitive scenarios where consistent and assertive behavioural strategies prevail.

Additionally, differences in larval ecology between the species help explain the observed behavioural patterns. *Polypedates cruciger* is a lentic, macrophagous nektonic tadpole, physiologically and behaviourally adapted to occupy the open water column where exploratory and active behaviours are advantageous (Altig and Johnston, 1989). *Duttaphrynus melanostictus*, on the other hand, typically occupies benthic or littoral zones, where environmental heterogeneity may select for behavioural flexibility rather than consistency. Thus, the repeatability and syndrome differences observed in this study are consistent with the ecological specialisations and competitive interactions documented for these species.

The results also have broader ecological and conservation implications. Amphibians are widely recognised as sensitive bio-indicators due to their permeable skin and reliance on aquatic environments (Burlibasa et al., 2011; Muller et al., 2015). Behavioural traits such as exploration, boldness, and activity influence exposure to pollutants, predation risk, and habitat quality. Proactive individuals may be more susceptible to contaminants due to increased movement and risk-taking, while reactive individuals may avoid polluted or dangerous microhabitats. Understanding baseline behavioural patterns can therefore improve detection of early behavioural disruptions caused by sub-lethal stressors such as pesticide runoff, heavy metal contamination, or altered hydrology.

Behavioural traits can also influence the success of conservation actions. For example, translocation programs or captive breeding initiatives may inadvertently select for particular behavioural types, potentially reducing behavioural diversity or altering fitness outcomes in the wild (Attard et al., 2016; Merick et al 2017). Endemic species like *P. cruciger* may be particularly susceptible to such effects. Integrating behavioural assessments into management planning can help ensure that reintroduced or captive-bred individuals possess behavioural profiles suited to surviving in natural habitats.

Despite its comprehensive scope, this study has several limitations that should be addressed in future research. First, the behavioural assays were conducted under controlled laboratory conditions, which, although necessary for standardisation, may not fully represent the complexity of natural habitats where predation cues, fluctuating resource availability, and variable microclimates can influence behavioural expression and repeatability. Incorporating semi-natural or in-situ behavioural trials would help validate whether the observed personality traits persist under ecologically realistic conditions. Second, while short- and long-term intervals were examined, a more continuous sampling regime across developmental stages would provide clearer insight into when behavioural traits stabilise or reorganise during ontogeny. Third, physiological or neuroendocrine correlates of behaviour were not measured; integrating metabolic rate, stress hormones, or sensory traits could clarify the mechanisms underpinning both species' behavioural strategies. Finally, the sample size, though adequate for mixed-effects modelling, may limit the detection of weaker behavioural syndromes. Expanding sampling across multiple populations and habitats would allow researchers to test how environmental gradients, genetic structure, and anthropogenic stressors shape behavioural consistency and covariance. Addressing these limitations will deepen our understanding of amphibian personality development and improve predictions of how behavioural variation interacts with ecological change and conservation management.

In conclusion, this study demonstrates that larval stages of *P. cruciger* and *D. melanostictus* express distinct personality traits and behavioural syndromes that differ in strength, stability, and organisation. *Polypedates cruciger* exhibited stable long-term behavioural architecture, while *D. melanostictus* showed high short-term consistency and long-term flexibility. These findings contribute to a deeper understanding of amphibian behavioural ecology, competition, and

developmental plasticity. Future research should explore how these behavioural patterns influence survival and metamorphic success, how they change across ontogeny, and how environmental disturbances modify behavioural consistency. Long-term field studies, cross-stage assessments, and ecological experiments under natural conditions will further clarify the role of personality in amphibian ecology and conservation.

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